Supplementary Material: MahimandalSR Source and Implementation Audits

S1. Source-product overlap

The audit reads the frozen reflectance16 paper release manifest (SHA-256):

`6bf97403ee40ccf0cbd38b2e6e4932e01973e5d00e211e0acfb1e0b810efc5e7`

It compares exact `landsat_id` and `sentinel_id` values and their pairs across the released train, validation, and test partitions. The `spatial_block_id` sets are disjoint, but the source-product sets are not. The final column counts patches in the second-named split whose exact Landsat–Sentinel-2 source-product pair also occurs in the first-named split. Source reuse does not establish pixel-level overlap between patches, but it precludes interpreting this as an unseen-acquisition evaluation.

Table S1: Exact source-product overlap between release partitions.

Splits	Blocks	Landsat	Sentinel-2	Source pairs	Second-split patches
train–test	0	55	64	68	381
train–val	0	56	64	65	335
val–test	0	25	24	24	90

The test set comprises 731 patches in 13 held-out 100-km blocks. The corresponding training and validation counts are 3,424 patches in 247 blocks and 645 patches in 14 blocks. Complete source-ID lists and affected-patch counts are stored in `generated/source_product_overlap.json`. The audit can be regenerated with `scripts/audit_source_products.py`; reassigning source-linked groups would define a new release and require retraining and re-evaluation.

S2. Implementation compatibility ledger

The main quantitative tables contain only completed benchmark runs that passed the recorded compatibility check. In Table S2, "failed" means the tested implementation did not pass that check; it does not imply that the model family cannot work on this dataset. "Unsupported" denotes a modality that was not verified for the registered implementation. The exported ledger does not retain causes for the failures, so no cause is inferred.

Table S2: Status of the 14 registered trainable candidates. Bicubic requires no trainable preflight and is not listed.

Model	RGB preflight	RGB test	RGB-NIR preflight	RGB-NIR test
EDSR	passed	reported	unsupported	–
RCAN	passed	reported	unsupported	–
RDN	passed	reported	passed	reported
SwinIR	failed	–	failed	–
HAT	failed	–	failed	–
OmniSR	failed	–	failed	–
SRVGG (pixel-only)	passed	reported	passed	reported
CAFRN	passed	reported	passed	reported
DCM	passed	reported	unsupported	–
FENet	passed	reported	unsupported	–
HAUNet	failed	–	unsupported	–
HSENet	failed	–	unsupported	–
MHAN	failed	–	unsupported	–
SAN	failed	–	unsupported	–

S3. Frozen-release split and cloud-boundary audit

The audit in `scripts/audit_release_quality.py` requires the frozen manifest digest above and counts both patches and distinct 100-km block IDs for each landscape stratum. All nine strata occur in each partition, but the rare categories have very few distinct test blocks, which may also share source products.

Table S3: Pairs and distinct spatial blocks (in parentheses) by landscape and split.

Landscape	Train	Validation	Test
Alpine	201 (24)	51 (2)	101 (3)
Cropland	756 (65)	108 (6)	136 (7)
Desert/bare	26 (12)	19 (2)	5 (1)
Forest	604 (101)	98 (7)	77 (6)
Mixed	122 (55)	21 (4)	19 (5)
Mountain	525 (109)	105 (6)	170 (6)
Shrub/grass	601 (102)	102 (7)	97 (5)
Urban	550 (81)	131 (8)	119 (6)
Water/wetland	39 (27)	10 (5)	7 (2)

The completed Kaggle audit opens all 9,600 rasters and transforms their header-derived footprint polygons to EPSG:6933. Minimum cross-split footprint distances are zero in all three comparisons. Using the audit's intersection-area threshold of $> 1\,\mathrm{m}^2$, 14 test patches overlap training and nine overlap validation, with one common patch (22 unique test patches). The executed split configuration used `guard_m=0`; no boundary guard was applied. These planar projected measurements do not constitute independently surveyed geodetic accuracy or a spatial buffer.

All 9,600 TIFF checksums and headers pass the audit, with no missing or orphan files. All rasters have all-valid raster masks and no explicit nodata value. These masks are not stored QA/SCL cloud masks. Original-source QA/SCL re-reading succeeds for all 4,800 pairs: mean obstruction is 0.12286%, maximum obstruction is 0.99740%, and all unrounded clear fractions exceed 0.99. The three rounded-boundary clear fractions are 0.99003472, 0.99003472, and 0.99002604.

Three source-window spot checks reproduce the stored red-band conversion without a pixel mismatch for either sensor. This verifies reproduction of the preparation code, including its Sentinel subtraction of 1,000 digital numbers. A separate metadata-only audit retrieves the original product XML for all 648 distinct Sentinel products (323 at processing baseline 05.11 and 325 at 05.12), covering every released pair. XML and STAC product URIs and baselines agree, and explicit XML physical-band identifiers map the required blue, green, red, and NIR offsets. All 648 products specify `BOA_QUANTIFICATION_VALUE`=10,000 and `BOA_ADD_OFFSET`=−1000 DN for B02/B03/B04/B08. Their formula $\rho = (DN + \text{offset})/\text{quantification}$ agrees with the preparation formula for all four bands, with no unresolved or disagreeing product.

The implementation is `scripts/audit_sentinel_product_metadata.py`; the `sentinel_product_metadata_audit/` evidence directory under `mahimandal-dataset/` contains original STAC and XML bytes, unsigned URLs, retrieval timestamps, SHA-256 receipts, per-product results, and a complete checksum inventory. It supports offline revalidation and does not download image tiles. Generic Rasterio scale/offset defaults are not used as physical coefficients. This closes product-coefficient agreement, not all-pixel correctness or an independent audit of provider asset harmonization.

S4. Acquisition timing by year and broad region

The following cross-tab is generated from the frozen release manifest with `scripts/build_review_temporal_table.py`. Regions are broad sampling strata, not administrative units. Empty year–month combinations are omitted; October has no retained pairs.

Table S4: Released pairs by Landsat UTC acquisition year, month, and sampling region. Each paired Sentinel-2 observation has the same UTC date.

Year	Month	Extreme north	Northern band	Rest of Bharat	Total
2025	03	0	17	0	17
2025	09	129	58	113	300
2025	11	0	2	0	2
2025	12	0	0	230	230
2026	01	39	29	2487	2555
2026	02	83	30	56	169
2026	03	7	20	218	245
2026	04	4	57	28	89
2026	05	89	176	341	606
2026	06	69	37	347	453
2026	07	62	2	13	77
2026	08	15	0	0	15
2026	09	32	10	0	42
Total	–	529	438	3833	4800

S5. Completed benchmark source provenance

The quantitative benchmark includes only the following pinned implementations and bicubic interpolation. Full revisions are in the benchmark registry; constructor, modality, checkpoint, and per-image test-score receipts accompany the result bundle. A registered candidate without a valid completed test run is in the compatibility ledger, not this table.

Table S5: Source revisions for completed learned-model evaluations. SRVGG stopped before adversarial training.

Evaluated model	Source repository	Commit prefix
EDSR	sanghyun-son/EDSR-PyTorch	`8dba5581a750`
RCAN	yulunzhang/RCAN	`f722e5d0e1ef`
RDN-B port	sanghyun-son/EDSR-PyTorch	`8dba5581a750`
CAFRN	lxy0922/CAFRN	`fef57e89c461`
DCM comparison implementation	Shaosifan/HSENet	`29e71e33a2d6`
FENet (paper identity unverified)	wangzheyuan-666/FeNet	`165954b920c0`
SRVGG 3× adaptation	xinntao/Real-ESRGAN	`a4abfb2979a7`

The three RGB benchmark figures use test pair IDs recorded in `generated/benchmark_gallery_provenance.json`. Their acquisition dates and within-pair time gaps are: urban, 16 May 2026 (13.0 minutes); cropland, 14 January 2026 (12.8 minutes); and mountain, 14 May 2026 (10.8 minutes). This full-paper gallery is distinct from the short-overview gallery and should not be described as the same sample set.

- Urban: `LC09_L2SP_147038_20260516_02_T1__43RFQ_20260516__8_33`.
- Cropland: `LC09_L2SP_141042_20260114_02_T1__44RRP_20260114__12_34`.
- Mountain: `LC09_L2SP_149037_20260514_02_T1__43SCT_20260514__9_17`.

The frozen release manifest maps each pair ID to its full Landsat and Sentinel-2 source product identifiers.

S6. Overlap-excluded benchmark sensitivity

The original 731-pair test set and its validation-selected models remain unchanged. We remove the union of test patches with positive-area footprint intersection ($> 1\,\mathrm{m}^2$ in EPSG:6933) against training or validation, using `nearest_cross_split_footprints.csv` from the completed audit. The remaining 709 patches occupy all 13 test blocks and all nine landscape strata. No model is retrained or selected again. Each archived score file is checked for unique pair IDs and exact membership of the frozen test set before filtering. Means are per-image averages. Paired PSNR differences use the corresponding archived bicubic predictions. Percentile intervals resample 13 blocks uniformly with replacement (2,000 draws, seed 42), pooling all paired differences in each draw. These intervals describe block-sensitive variation, not independent source-scene or training-seed replication. The retained set still includes 369 patches whose exact source pair occurs in training and is therefore not source-disjoint or buffered.

Table S6: Post hoc overlap-excluded sensitivity for all completed fixed-model runs. PSNR and paired gain over bicubic are in dB. SRVGG is pixel-only. The full 731-pair benchmark remains the primary analysis.

Output	Model	PSNR (731)	PSNR (709)	Δ_{709}	95% block CI
RGB	CAFRN	36.922	36.899	+1.525	[1.058, 2.062]
RGB	DCM	36.796	36.768	+1.394	[0.703, 2.128]
RGB	EDSR	36.763	36.755	+1.381	[0.630, 2.062]
RGB	FENet	36.644	36.622	+1.248	[0.612, 1.991]
RGB	RCAN	36.595	36.568	+1.194	[0.539, 1.929]
RGB	RDN	37.669	37.640	+2.266	[1.575, 3.078]
RGB	SRVGG (pixel-only)	35.644	35.622	+0.247	[-0.242, 0.887]
Four-band	CAFRN	34.864	34.861	+0.971	[0.483, 1.537]
Four-band	RDN	36.151	36.147	+2.257	[1.685, 2.875]
Four-band	SRVGG (pixel-only)	33.909	33.902	+0.012	[-0.271, 0.357]

The reproducible implementation is supplied as:

`benchmarks/mahimandal160/overlap_sensitivity.py`

Its output contains the 22 excluded IDs, 709 retained manifest rows, per-class counts, all available archived metrics, paired block-bootstrap results, input hashes, and the table source. Geometry and score inputs are read only; raw imagery and trained checkpoints are not modified.

S7. Spatially held-out feature-match diagnostic

The fixed seed-42 sample contains 100 pairs: 11 per landscape class, except cropland with 12; 64 training, 18 validation, and 18 test pairs. Within each stratum, sampling prefers previously unrepresented blocks and acquisition pairs before deterministic hash tie-breaking. The selected sample spans 90 blocks and 99 source pairs. It is not selected using match quality or any model result. Pair IDs and input TIFF hashes are retained.

Sentinel-2 is reduced by nonoverlapping 3×3 area means to the 30-m grid. SIFT features use separately percentile-stretched visible luminance for detection only, mutual two-nearest-neighbour ratio matching at 0.75, and removal of repeated half-pixel feature locations. No displacement gate is applied. Occupied cells of a 4×4 source-image grid are hash-shuffled; one third (rounded up) are held out before transform fitting. A similarity transform uses training matches only, with RANSAC threshold 1.5 Landsat pixels. Evaluable pairs require eight fitting matches, six held-out matches, and at least six training inliers representing half the training matches. No held-out residual is used to select inliers or drop a case.

Ninety-eight pairs are evaluable; two alpine pairs fail the predefined match count requirement. Mean per-pair held-out RMS displacement before fitting is 0.692 Landsat pixels; fitted-transform residual RMS is 0.677. The mean per-pair median displacement before fitting is 0.253 Landsat pixels. One Landsat pixel equals three Sentinel target pixels or 30 m on the declared grid. Gross held-out mismatches are retained, so RMS values are sensitive to descriptor outliers. Evaluable-only statistics do not describe all 100 pairs or establish release-wide positional accuracy.

Table S7: Mean per-pair held-out RMS descriptor displacement and fitted-transform residual in *Landsat pixels*, computed only for evaluable pairs. No statistic is a ground-control accuracy estimate.

Stratum	Attempted	Evaluable	Identity RMS	Fitted residual RMS
alpine	11	9	1.609	1.610
cropland	12	12	0.465	0.450
desert/bare	11	11	0.462	0.439
forest	11	11	0.662	0.653
mixed	11	11	0.510	0.505
mountain	11	11	0.935	0.903
shrub/grass	11	11	0.754	0.760
urban	11	11	0.495	0.463
water/wetland	11	11	0.524	0.502

The code `scripts/audit_registration_holdout.py` exports every match, its fit/held-out assignment, coordinates in both grids, image hashes, fitted transforms, failures, and 100 inspection panels. An offline inspection gallery and blank review CSVs are supplied. Zero pairs are recorded as manually reviewed; automatic correspondence consistency is not independent surveyed ground truth. The release imagery and benchmark scores are unchanged.

S8. Older-record auxiliary provenance recovery

The older preparation tree contains the same 4,800 manifest pair IDs as the frozen release. The compared source identifiers, landscape labels, nine Esri percentages, annual/service fields, DEM source descriptor and saved terrain quantiles agree without a difference. Reapplying the recorded Esri-plus-terrain rules to saved rounded metadata reproduces all 4,800 landscape labels. This demonstrates rule reproducibility, not independent land-cover classification accuracy.

The read-only recovery audit inspects all 9,606 older TIFF headers, including domain and band tags, and inventories the available text/JSON/CSV metadata. No explicit auxiliary tile/version lineage is found in the TIFF tags. The additional six rasters are older staging files, not additions to the frozen release. Original per-patch Esri raster IDs and DEM item/version receipts remain unestablished; current provider metadata cannot reconstruct those missing historical records as fact. The older protocol's target split fractions do not replace the frozen realized block assignments. The implementation is `scripts/audit_legacy_auxiliary_lineage.py`.

S9. Reader radiometry, identified fill, and exclusion sensitivity

The source-reader audit fixes two pairs by SHA-256 ordering of frozen pair IDs, requiring distinct Sentinel products. All four bands per sensor reproduce the release without a mismatch (16 windows; 2,048,000 values). Both examples use Landsat 8 and contain no detected invalid locations; they are not a platform-stratified all-release sample. A synthetic GeoTIFF with non-default scale/offset tags separately confirms raw-DN behavior of the installed reader. It is a unit test, not an original-source measurement.

Stored zeros occur in 1,046 pair IDs: 667 Landsat and 520 Sentinel rasters. All 1,187 candidate rasters were checked against original windows for each band containing zeros. The conversion reproduces without mismatches. The audit records 29 Landsat rasters with 148 identified invalid locations and 381 Sentinel rasters with 4,540 such locations; 404 distinct pairs are affected. Most zero band values instead come from valid clipping: 307,152 Landsat and 2,292,305 Sentinel band values. Band-value counts must not be confused with unique spatial locations. Some Sentinel COGs omit a TIFF nodata header; their original product XML identifies nodata DN zero, consistent with the collector's explicit VRT setting. The first audit stopped on that missing-header convention; a separately versioned continuation retains successful records and verifies the XML instead of treating absent header metadata as a different fill value.

This audit concerns output-grid source nodata/read masks and DN zero. It does not certify saturation, clouds, original invalid interpolation-contributor weights, or every nonzero band value. Separate identified-fill masks are audit artifacts, not silently substituted benchmark masks. A mask value of one means identified fill; zero means no fill identified by this audit, not universal physical validity. The original release remains unchanged.

The affected split counts are 266 train, 58 validation and 80 test. Whole-pair exclusion yields 651 test pairs; adding the 22 footprint-overlap exclusions yields 629. Both retain nine strata and 13 blocks. No model is retrained and no prediction is recomputed. The original trained weights may have encountered identified fill; exclusion from testing does not remove that training history.

Table S8: Archived-score sensitivity to excluding the 80 test pairs with identified source fill. All models use the same retained 651 pairs. The last column is the paired gain over bicubic on those pairs, not over full-test bicubic. The registered SRVGG rows denote pixel-only training.

Output	Model	PSNR full	PSNR fill-excluded	Δ vs bicubic
RGB	CAFRN	36.922	37.478	+1.576
RGB	DCM	36.796	37.374	+1.472
RGB	EDSR	36.763	37.321	+1.418
RGB	FENet	36.644	37.205	+1.303
RGB	RCAN	36.595	37.117	+1.214
RGB	RDN	37.669	38.191	+2.289
RGB	RealESRGAN_SRVGG_X3	35.644	36.217	+0.315
RGB	bicubic	35.378	35.902	+0.000
RGBNIR	CAFRN	34.864	35.517	+1.013
RGBNIR	RDN	36.151	36.741	+2.237
RGBNIR	RealESRGAN_SRVGG_X3	33.909	34.526	+0.021
RGBNIR	bicubic	33.892	34.504	+0.000

The supplied audit and sensitivity scripts accompany machine-readable source/reader headers, numeric examples, XML/STAC hashes, resume receipts, candidate membership, completed-model metrics, class counts and 2,000-draw paired block-bootstrap intervals (seed 42).

S10. OLI2MSI channel and radiometric provenance

The local full-test audit verifies 100 matched identifiers, 200 float32 TIFFs, three channels, 160/480 geometry, finite values in $[0, 1]$, and red/green/blue TIFF interpretation tags. Band descriptions are absent, but interpretation tags and the publisher's 4/3/2 extraction configuration support the RGB mapping. No extra division by 10,000, gamma, display stretch or 8-bit quantization is applied in the reported transfer experiment.

The publisher repository is pinned at `da443af38994a0cbcec14eda62c87185764434a4`; the retained extraction and utility sources invoke TOA readers for both sensors. Mahimandal surface reflectance and OLI2MSI TOA reflectance share a numerical scale, not a verified common atmospheric correction. This difference is a provenance observation, not an isolated explanation for the measured rank changes. All 100 pairs have unequal LR/HR header bounds within the same CRS. Header-coordinate differences are not positional accuracy measurements, and official image arrays are not rewarped in this analysis. Authenticating the local repack against the original publisher archive would require a corresponding authoritative pixel inventory.

`scripts/audit_oli_preprocessing.py` retains the pinned publisher sources, local file hashes, RGB interpretation fields, per-file ranges, coordinate-bound differences and the archived transfer-cohort comparison. No new training or inference is involved.